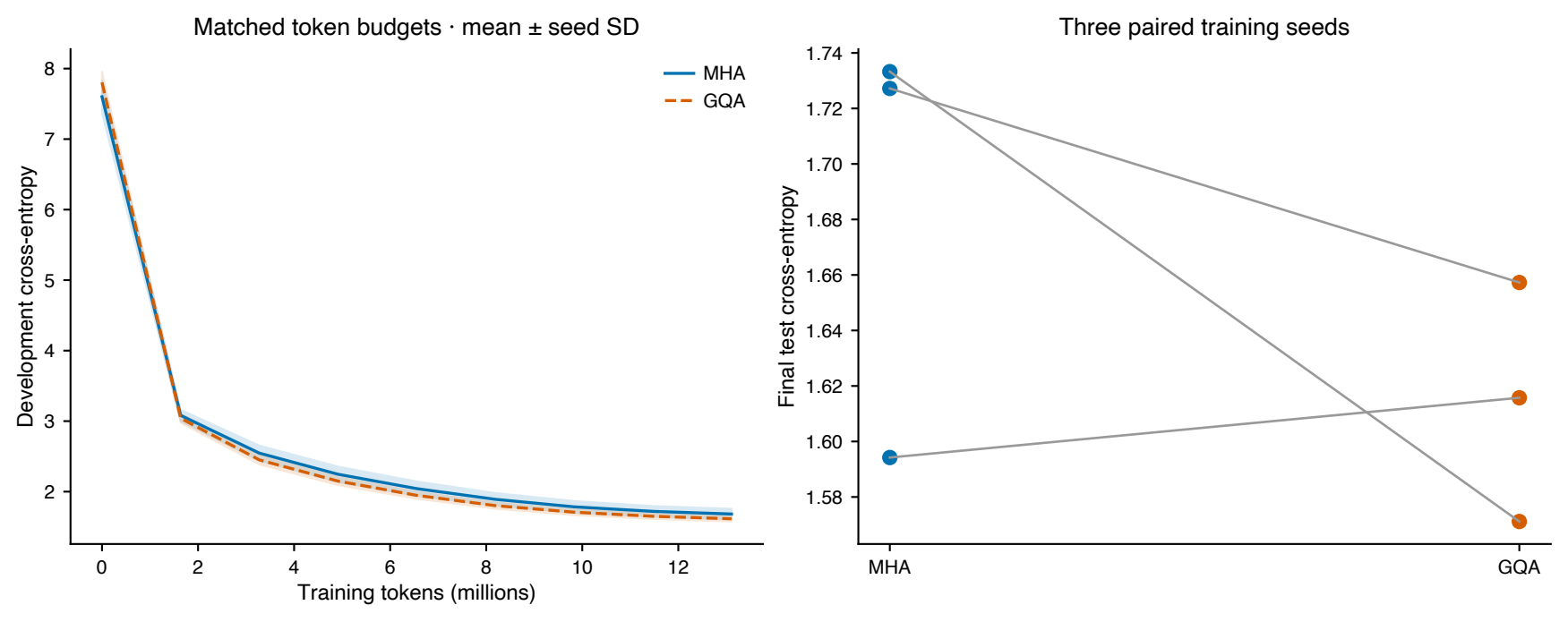
Matched token budgets · mean ± seed SD
MHA
GQA
Development cross-entropy
8
7
6
5
4
3
2
0
2
4
6
8
10
12
Training tokens (millions)
Three paired training seeds
Final test cross-entropy
1.74
1.72
1.70
1.68
1.66
1.64
1.62
1.60
1.58
MHA
GQA